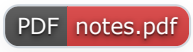
Notes_25.12: What is Validation Data?







🌐 **Language: English** | हिंदी / Hinglish | ⬅️ Lecture 25 Overview

---

📌 Executive Summary

Train-Validation-Test 3-way split, hyperparameter tuning without test set contamination.

🎯 Mathematical Formulations

Three-Way Data Partition Architecture:

$$ \mathcal{D} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{test}}, \quad \mathcal{D}_{\text{train}} \cap \mathcal{D}_{\text{val}} = \emptyset, ; \mathcal{D}_{\text{val}} \cap \mathcal{D}_{\text{test}} = \emptyset $$

Standard Split Ratios:

$$\text{Train: } 60 - 70$$

💡 Conceptual Principles & Architecture

Test Set Ko Chhuna Mana Hai!

- **Training Data:** Model ke weights aur parameters ko train karne ke liye.
- **Validation Data:** Hyperparameters tune karne ke liye (jaise kNN mein best K choose karna, Decision tree ki depth tune karna).

- **Test Data:** Final unbiased evaluation report card. Test data par kabhi koi parameter ya hyperparameter tune nahi kiya jata!

💻 Python Production Implementation

```
from sklearn.model_selection import train_test_split
import numpy as np

X = np.random.randn(1000, 10)
y = np.random.randint(0, 2, 1000)

# Step 1: Split into Train (70%) and Temp (30%)
X_train, X_temp, y_train, y_temp = train_test_split(X, y, test_size=0.30, random_

# Step 2: Split Temp into Validation (15%) and Test (15%)
X_val, X_test, y_val, y_test = train_test_split(X_temp, y_temp, test_size=0.50, r

print(f"Train Shape: {X_train.shape}, Val Shape: {X_val.shape}, Test Shape: {X_te
```